GLOBAL EDITION

MICROECONOMICS

Indifference curve

↓

each preference gives you the same level of satisfaction

* budget equation

14th Edition

P252

Q's 1-5 & 7

Michael Parkin



9 POSSIBILITIES, PREFERENCES, AND CHOICES

After studying this chapter, you will be able to:

- Describe a household's budget line and show how it changes when prices or income change
- Use indifference curves to map preferences and explain the principle of diminishing marginal rate of substitution
- Predict the effects of changes in prices and income on consumption choices

Copyright © 2023 Pearson Education, Ltd.

Consumption Possibilities

A household's consumption choices are constrained by its income and the prices of the goods and services available.

The **budget line** describes the limits to the household's consumption choices.

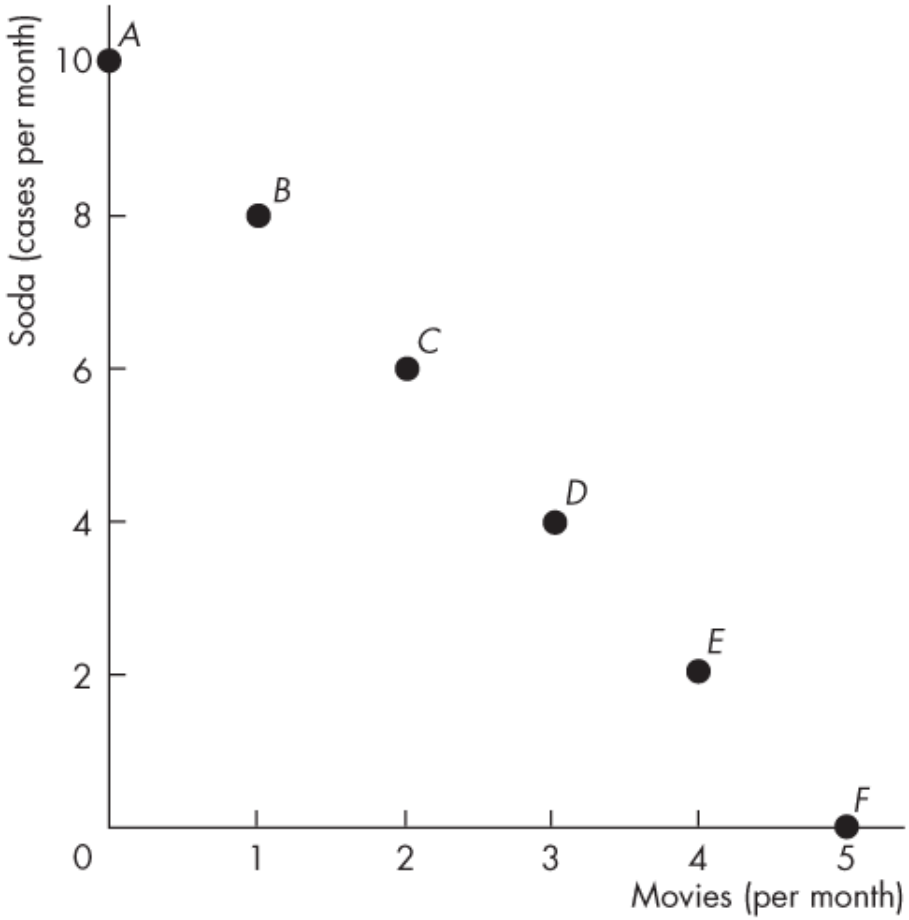
Copyright © 2023 Pearson Education, Ltd.

Consumption Possibilities

Budget Line

Lisa has $40 to spend, the price of a movie is $8, and the price of soda is $4 a case.

Consumption possibility	Movies (per month)	Soda (cases per month)
A	0	10
B	1	8
C	2	6
D	3	4
E	4	2
F	5	0



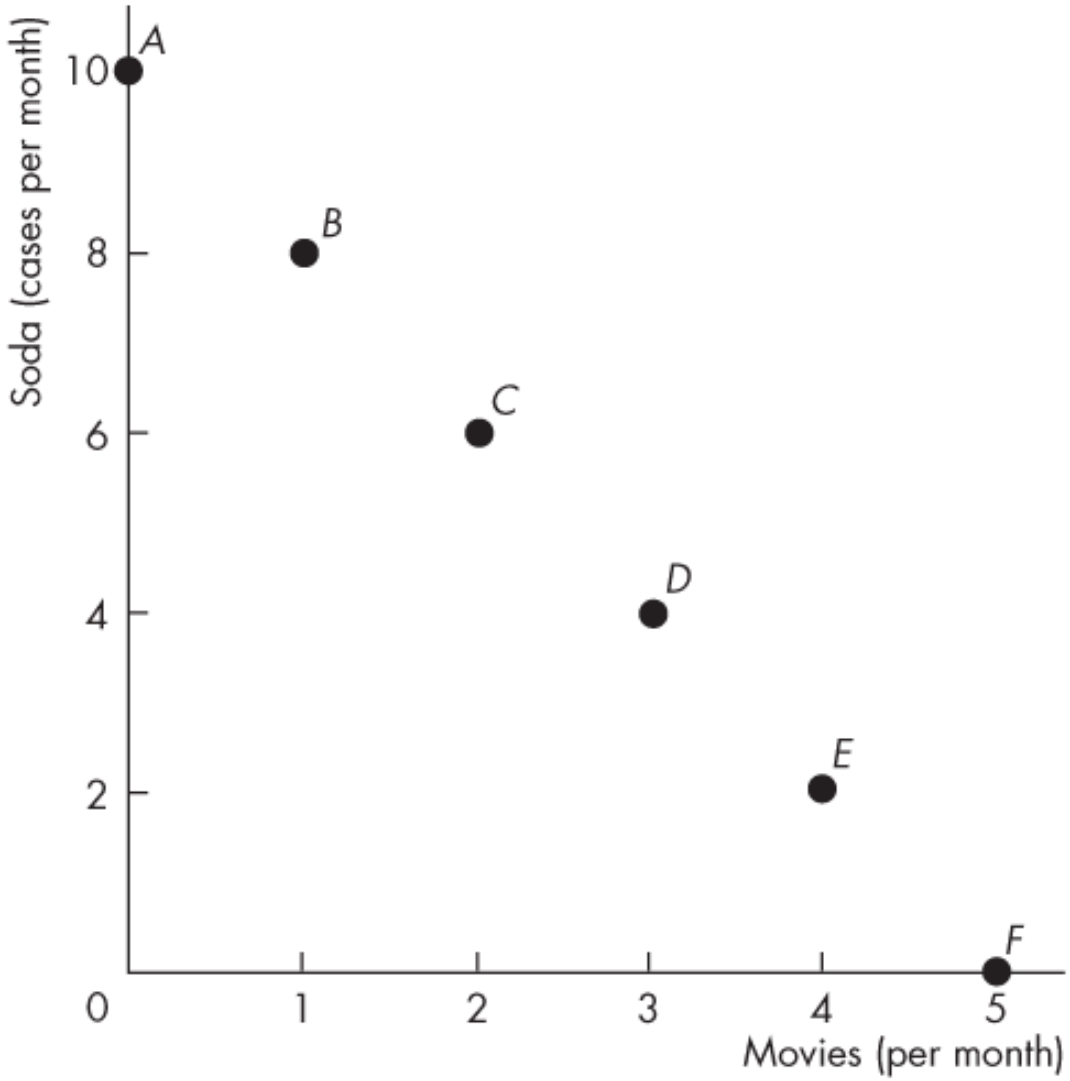
Copyright © 2023 Pearson Education, Ltd.

Consumption Possibilities

Lisa can afford any of the combinations at points *A* to *F*.

Some goods are indivisible goods and must be bought in whole units at the points marked.

Movies are indivisible goods.



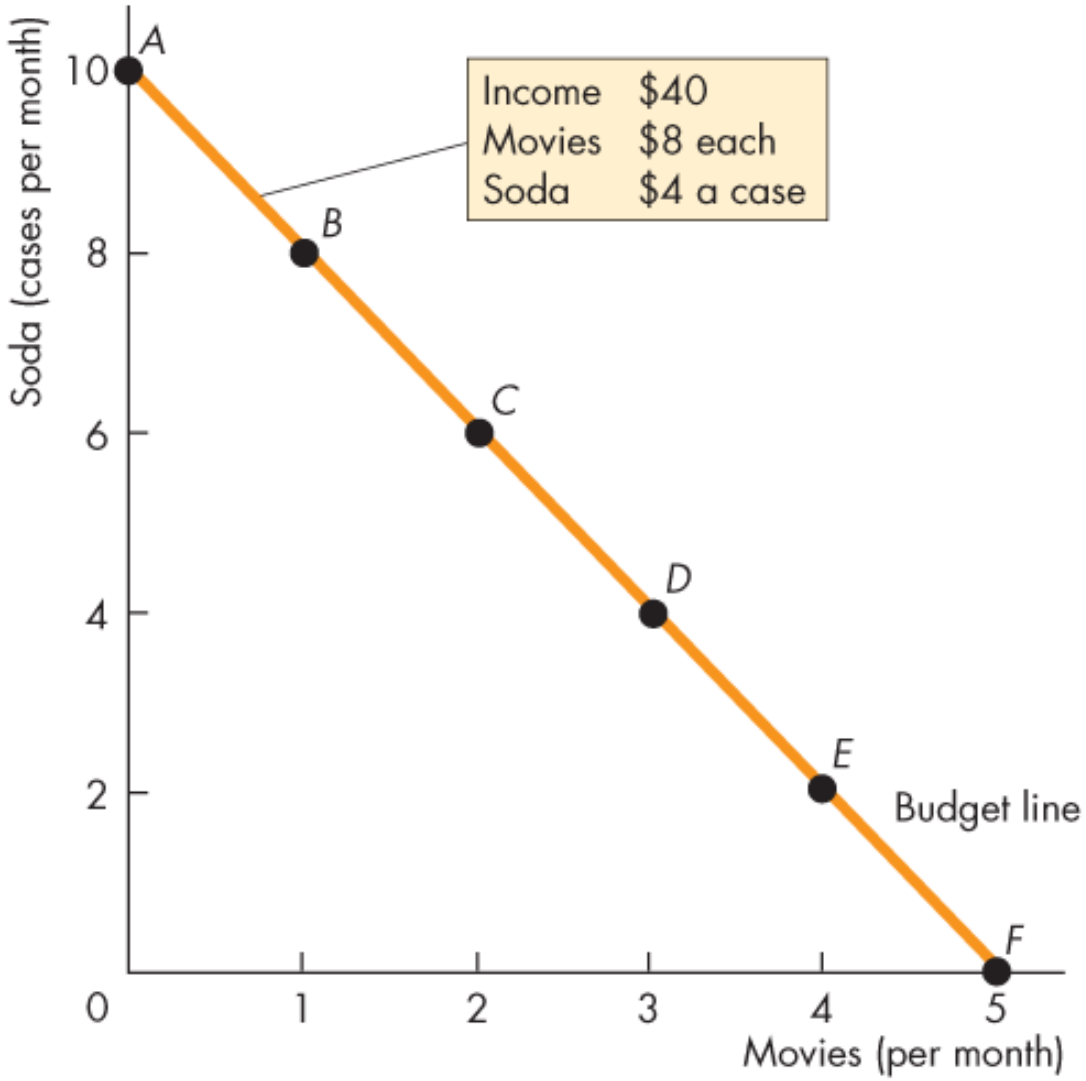
Copyright © 2023 Pearson Education, Ltd.

Consumption Possibilities

Other goods are divisible goods and can be bought in any quantity.

Gasoline is a divisible good.

The line through points *A* to *F* is Lisa's budget line.



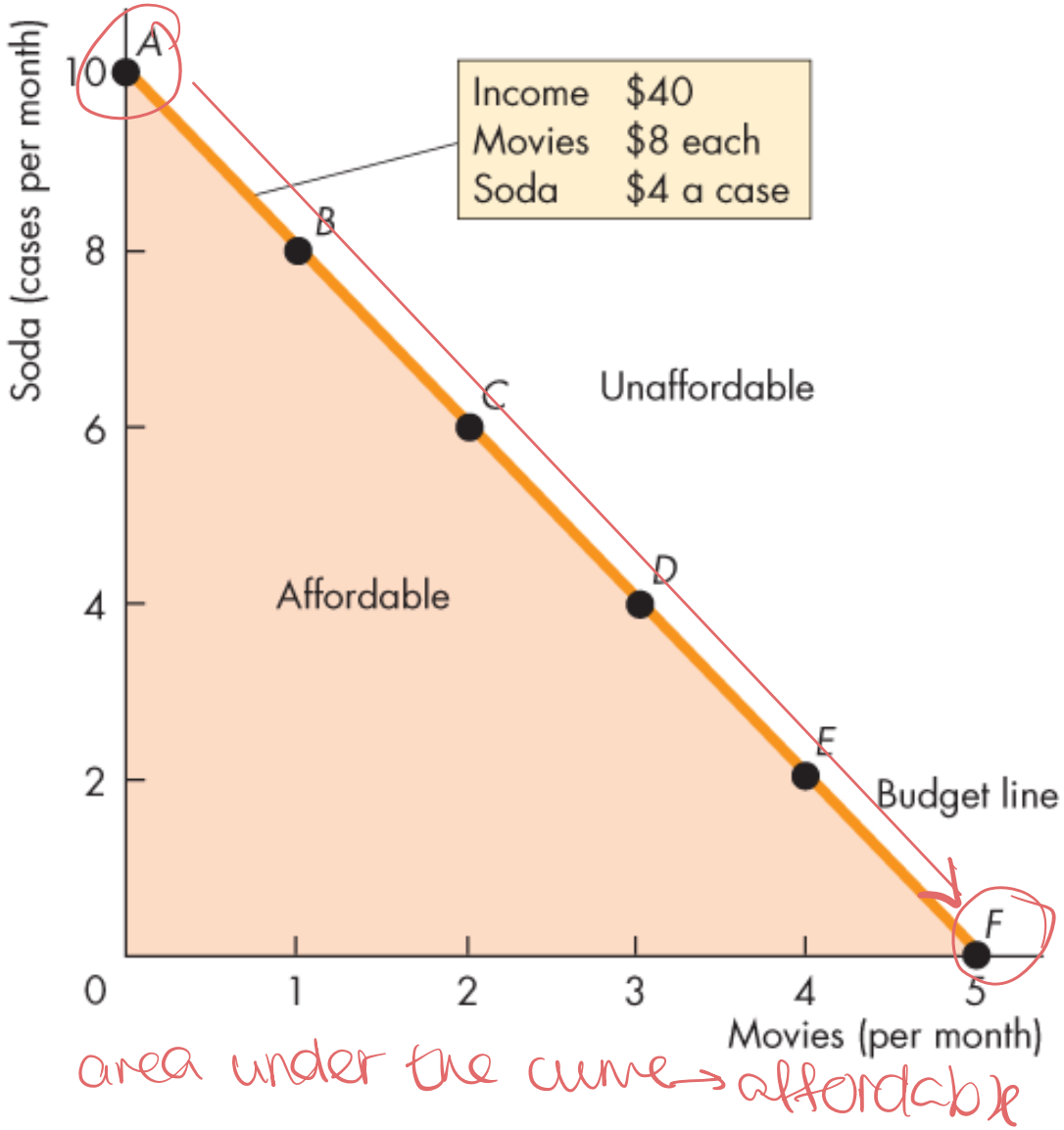
Copyright © 2023 Pearson Education, Ltd.

Consumption Possibilities

The budget line is a constraint on Lisa's consumption choices.

Lisa can afford any point on her budget line or inside it.

Lisa cannot afford any point outside her budget line.



$Y = P_M Q_M + P_S Q_S$

$40 = 4 Q_M + 4 Q_S$

$40 = 4 Q_M + 4(0)$

area under the curve → affordable

Copyright © 2023 Pearson Education, Ltd.

$\frac{40}{4} = \frac{4Q_M}{4}$ → $Q_M = 10$ → rotates slope outward → increases from 5 → 10 quantities

Consumption Possibilities

The Budget Equation

We can describe the budget line by using a budget equation.

The budget equation states that

↑ day-to-day expenses

$$\text{Expenditure} = \text{Income}$$

→ if you spend 100 to the seller, so they receive 100 in turn

Call the price of soda P_S, the quantity of soda Q_S, the price of a movie P_M, the quantity of movies Q_M, and income Y.

Lisa's budget equation is:

↓ income

$$P_S Q_S + P_M Q_M = Y.$$

Product 1 Product 2

hence why Expenses = Income

Copyright © 2023 Pearson Education, Ltd.

Consumption Possibilities

$$P_S Q_S + P_M Q_M = Y$$

Divide both sides of this equation by P_S, to give:

$$Q_S + (P_M/P_S)Q_M = Y/P_S$$

Then subtract $(P_M/P_S)Q_M$ from both sides of the equation to give:

$$Q_S = Y/P_S - (P_M/P_S)Q_M$$

Y/P_S is Lisa's **real income** in terms of soda.

P_M/P_S is the **relative price** of a movie in terms of soda.

Copyright © 2023 Pearson Education, Ltd.

Consumption Possibilities

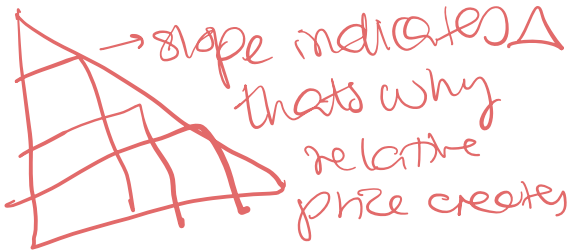
A household's **real income** is the income expressed as a quantity of goods the household can afford to buy.

Lisa's *real income* in terms of soda is the point on her budget line where it meets the *y*-axis.

A **relative price** is the price of one good divided by the price of another good.

Relative price is the magnitude of the slope of the budget line.

The relative price shows how many cases of soda must be forgone to see an additional movie.



Copyright © 2023 Pearson Education, Ltd.

if Lisa has $20, soda is $2 per cost
so her real income is income/price = $20/$2 = 10

a slope

Consumption Possibilities

$\frac{P_M}{P_S} \rightarrow \frac{\$6}{\$2} = 3$

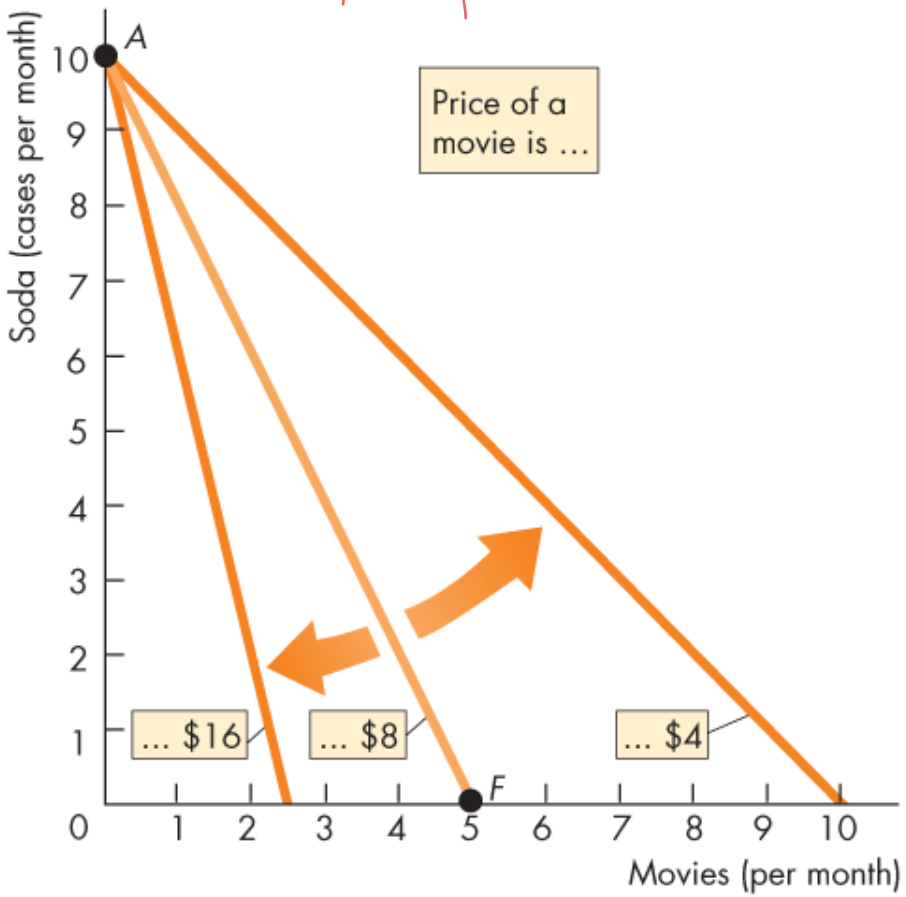
rotation

high price → shift inward
low price → shift outward

A Change in Prices

A change in the price of the good on the *x*-axis changes the *slope* of the budget line.

Figure 9.2(a) shows the rotation of a budget line after a change in the relative price of movies.



(a) A change in price

what happens/why budget line rotate?
bc of change in price

Copyright © 2023 Pearson Education, Ltd.

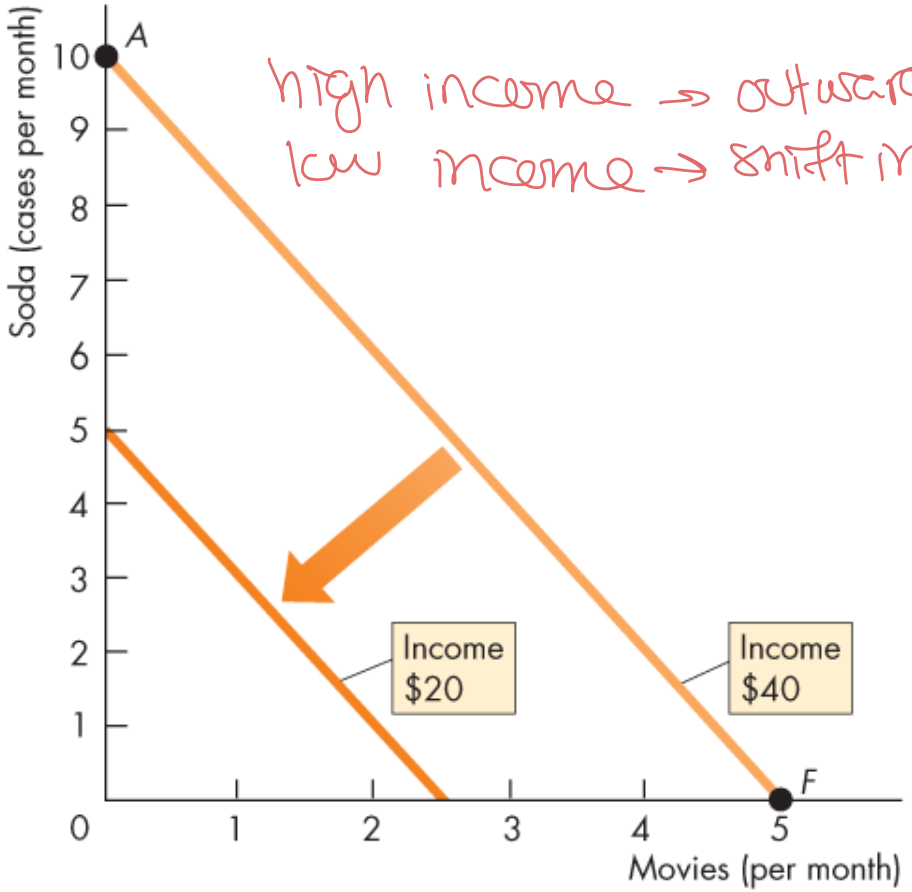
Consumption Possibilities

A Change in Income

An change in money income brings a parallel shift of the budget line.

The slope of the budget line doesn't change because the relative price doesn't change.

Figure 9.2(b) shows the effect of a fall in income.



(b) A change in income

Copyright © 2023 Pearson Education, Ltd.

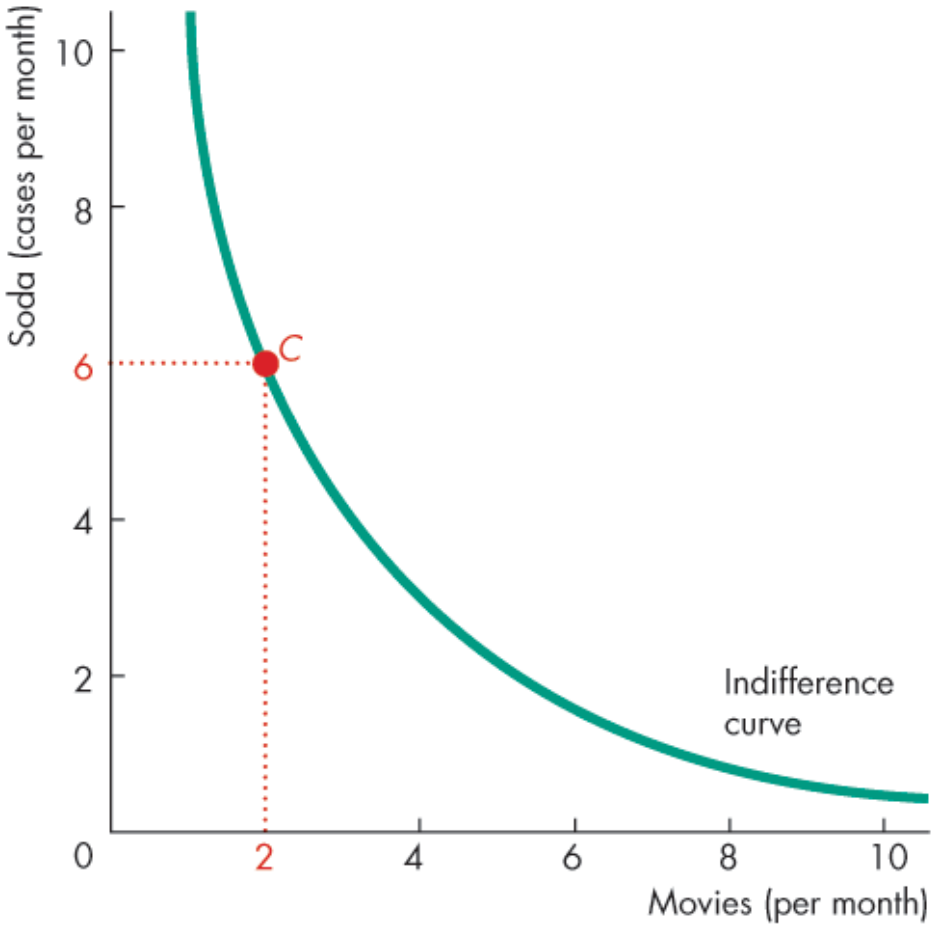
Preferences and Indifference Curves

We can represent Lisa's preference as a map.

An **indifference curve** is a line that shows combinations of goods among which a consumer is *indifferent*.

At point *C*, Lisa sees 2 movies and drinks 6 cases of soda a month.

Figure 9.3(a) illustrates Lisa's indifference curve.



(a) An indifference curve

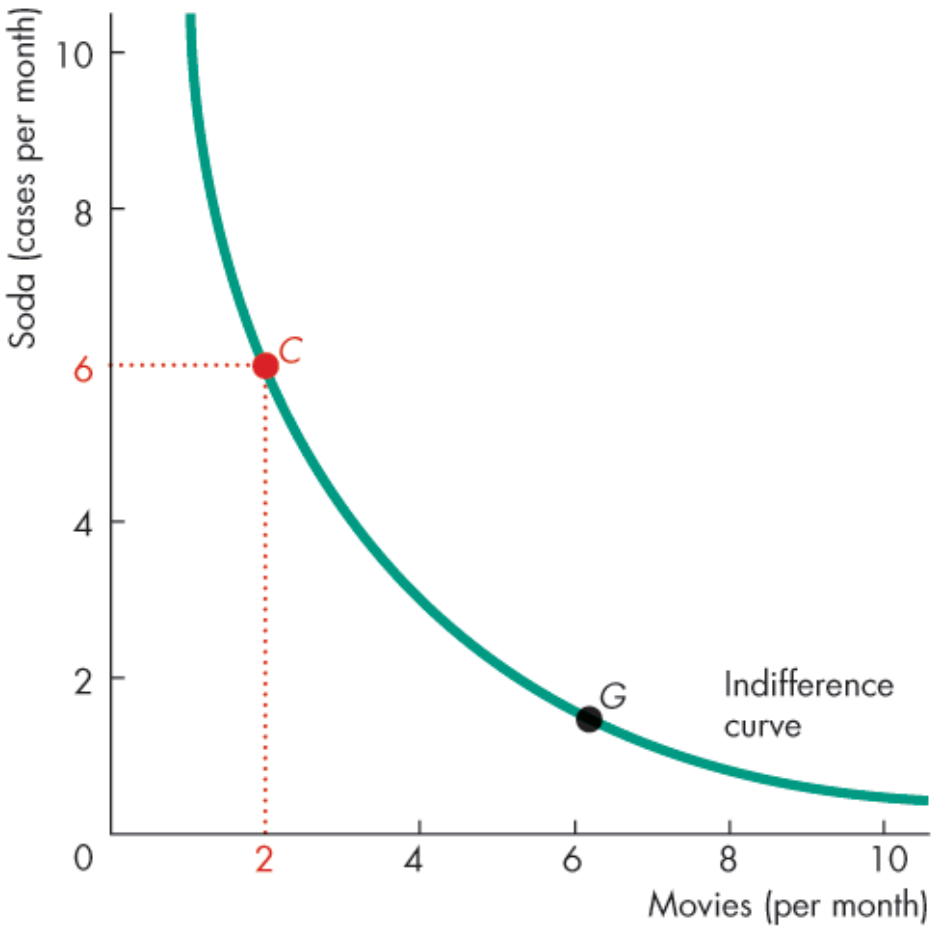
Copyright © 2023 Pearson Education, Ltd.

Preferences and Indifference Curves

Lisa can sort all possible combinations of goods into three groups: preferred to *C*, not preferred to *C*, and just as good as *C*.

An indifference curve joins all those points that Lisa says are just as good as *C*.

G is such a point. Lisa is indifferent between *C* and *G*.



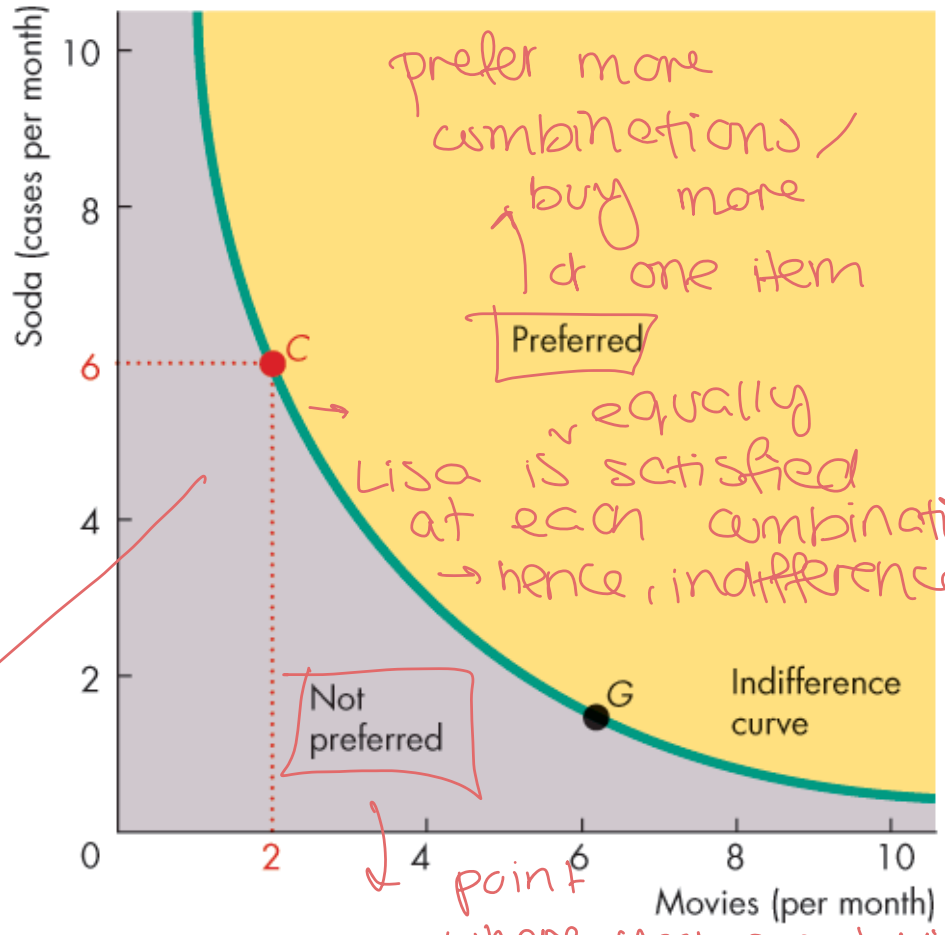
(a) An indifference curve

Copyright © 2023 Pearson Education, Ltd.

Preferences and Indifference Curves

Lisa prefers any point on the indifference curve to any point below the indifference curve.

Lisa prefers any point above the indifference curve to any point on the curve.



(a) An indifference curve

prefer more combinations / buy more of one item

equally

Lisa is satisfied at each combination → hence, indifference

downward slope

when you buy more of one good, you buy less from another

point where you are buying less

whatever combination you choose you aren't indifferent towards it - they provide equal satisfaction

Copyright © 2023 Pearson Education, Ltd.

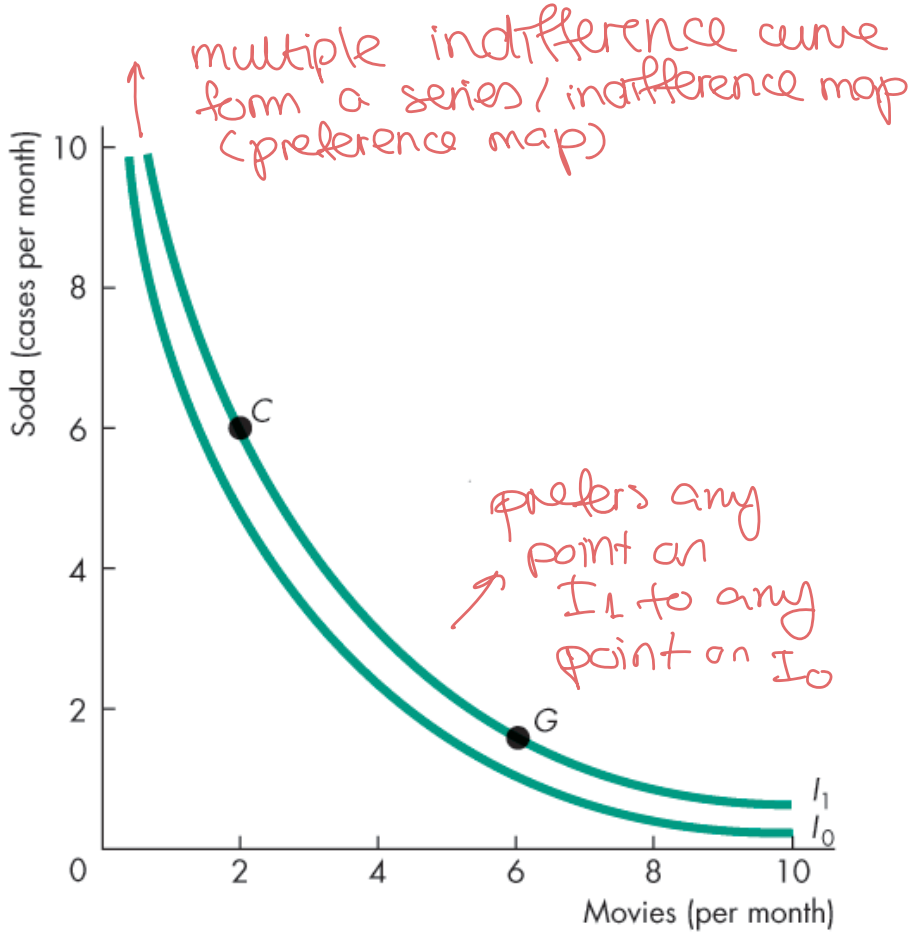
Preferences and Indifference Curves

A preference map is a series of indifference curves.

Call the indifference curve that we've just seen I_1.

I_0 is an indifference curve below I_1.

Lisa prefers any point on I_1 to any point on I_0 .



(b) Lisa's preference map

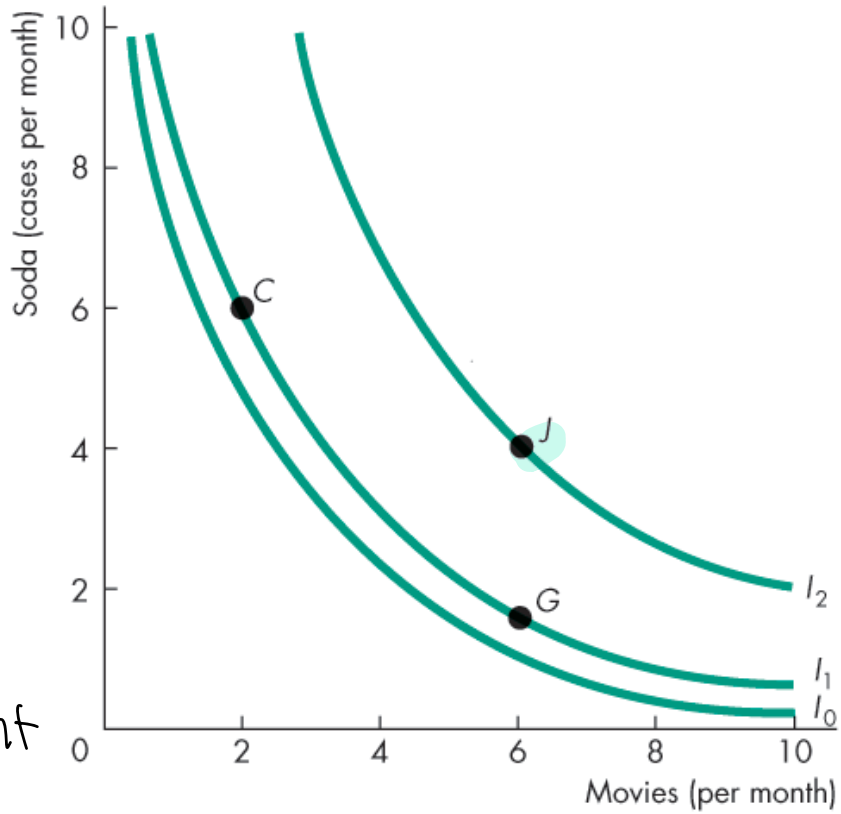
Copyright © 2023 Pearson Education, Ltd.

Preferences and Indifference Curves

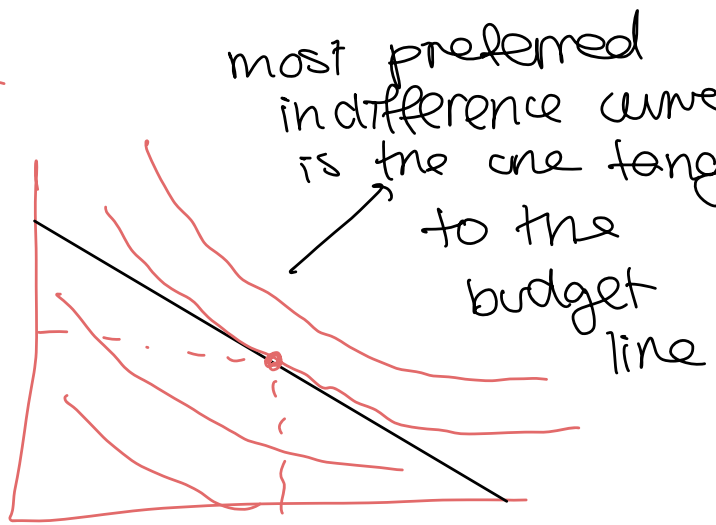
I_2 is an indifference curve above I_1.

Lisa prefers any point on I_2 to any point on I_1.

For example, Lisa prefers point J to either point C or point G.



(b) Lisa's preference map



Copyright © 2023 Pearson Education, Ltd.

Preferences and Indifference Curves

Marginal Rate of Substitution

The **marginal rate of substitution**, (*MRS*) measures the rate at which a person is willing to give up good *y* to get an additional unit of good *x*, while at the same time remaining indifferent (remaining on the same indifference curve).

The magnitude of the slope of the indifference curve measures the marginal rate of substitution.

Copyright © 2023 Pearson Education, Ltd.

Preferences and Indifference Curves

- If the indifference curve is relatively *steep*, *MRS* is high.

  In this case, the person is willing to give up a large quantity of *y* to get a bit more *x*.

- If the indifference curve is relatively *flat*, *MRS* is low.

  In this case, the person is willing to give up a small quantity of *y* to get more *x*.

Copyright © 2023 Pearson Education, Ltd.

Preferences and Indifference Curves

A diminishing marginal rate of substitution is the key assumption of consumer theory.

A **diminishing marginal rate of substitution** is a general tendency for a person to be willing to give up less of good *y* to get one more unit of good *x*, while at the same time remaining indifferent as the quantity of good *x* increases.

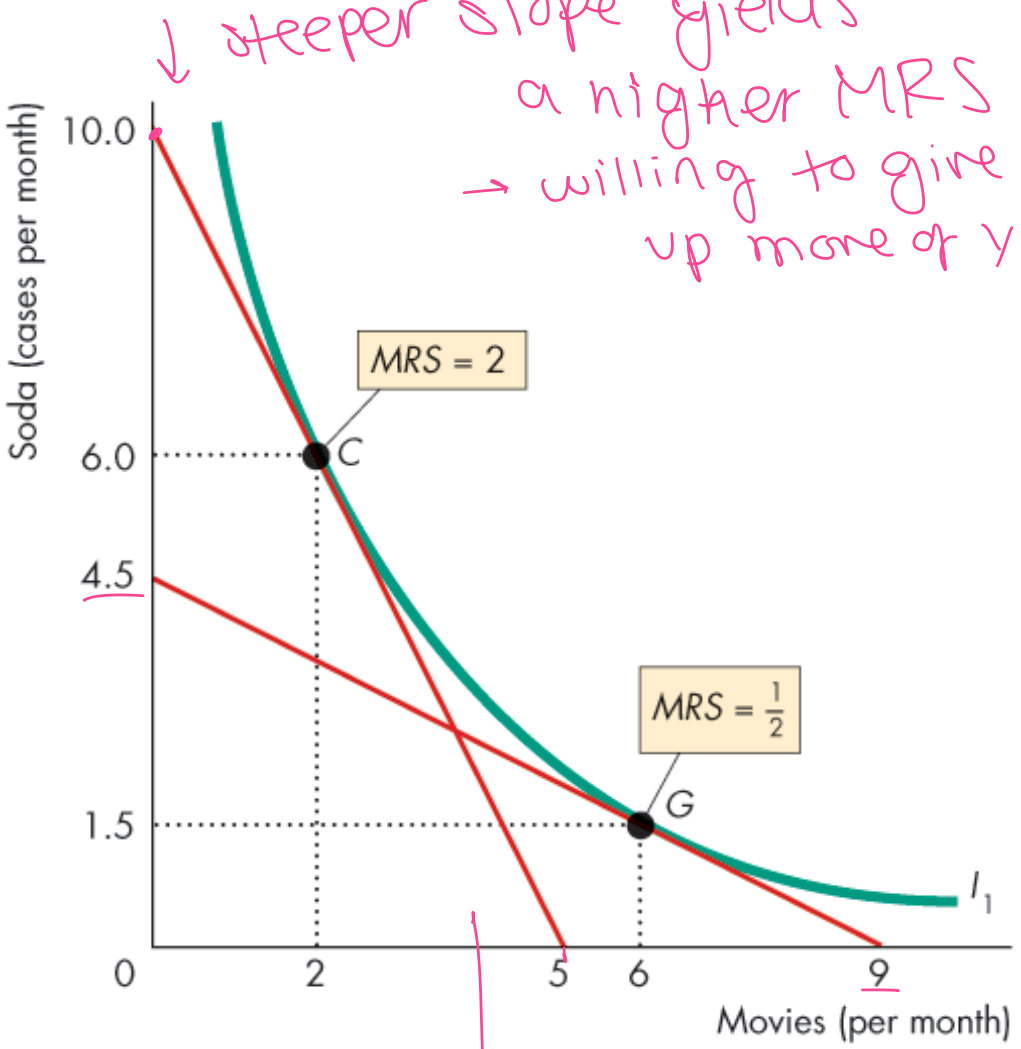
Copyright © 2023 Pearson Education, Ltd.

Preferences and Indifference Curves

Figure 9.4 shows the diminishing *MRS* of movies for soda.

At point *C*, Lisa is willing to give up 10 cases of soda to see 5 more movies—her *MRS* is 2.

At point *G*, Lisa is willing to give up 4.5 cases of soda to see 9 more movies—her *MRS* is ½.



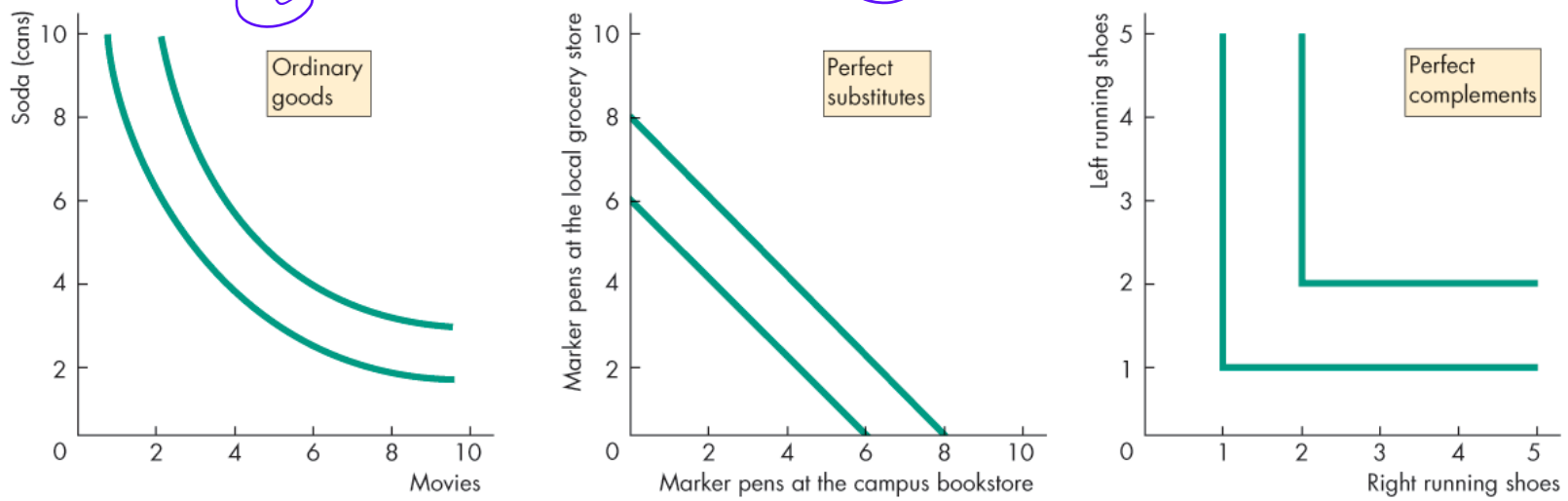
Copyright © 2023 Pearson Education, Ltd.

Preferences and Indifference Curves

Degree of Substitutability

The shape of the indifference curves reveals the degree of substitutability between two goods.

Figure 9.5 shows the indifference curves for ordinary goods, perfect substitutes, and perfect complements.



(a) Ordinary goods (b) Perfect substitutes (c) Perfect complements

Copyright © 2023 Pearson Education, Ltd.

Predicting Consumer Choices

Best Affordable Choice

The consumer's best affordable choice

- Is on the budget line.
- Is on the highest attainable indifference curve.
- Has marginal rate of substitution equal to relative price.

Copyright © 2023 Pearson Education, Ltd.

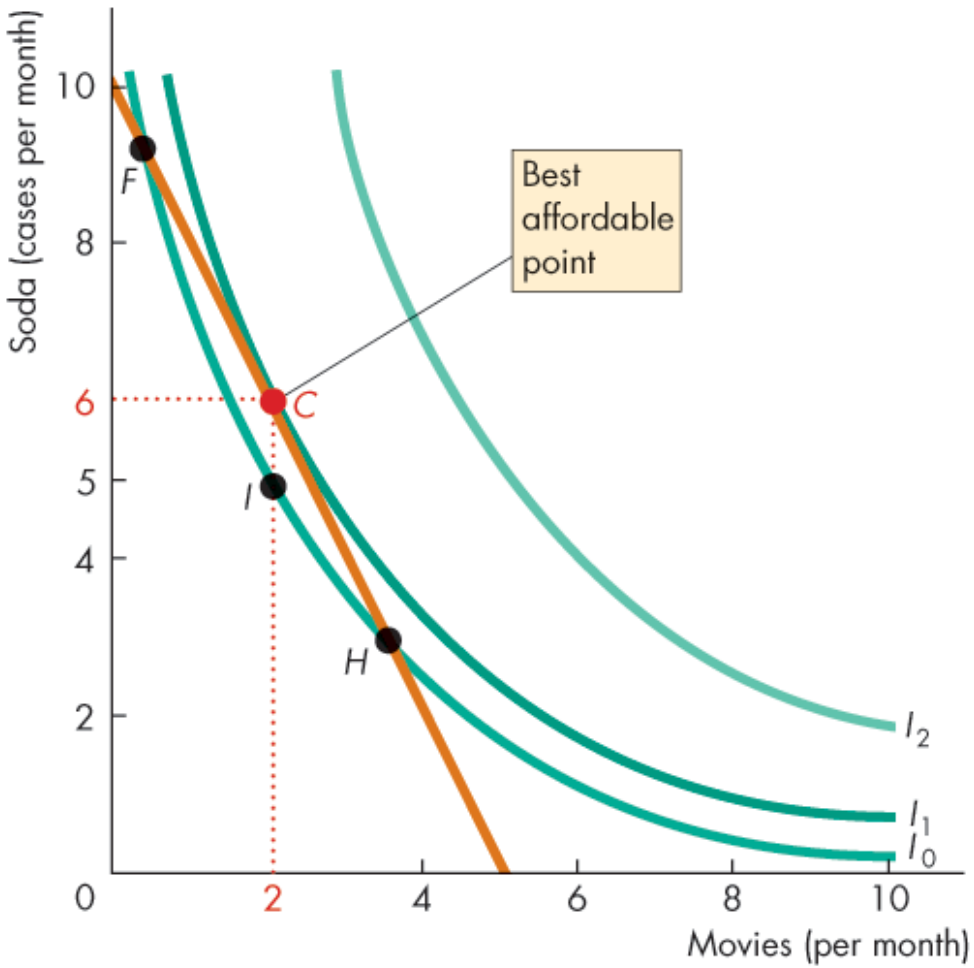
Predicting Consumer Choices

Here, the best affordable point is *C*.

Lisa can afford to consume more soda and see fewer movies at point *F*.

And she can afford to see more movies and consume less soda at point *H*.

But she is indifferent between *F*, *I*, and *H* and she prefers *C* to *I*.



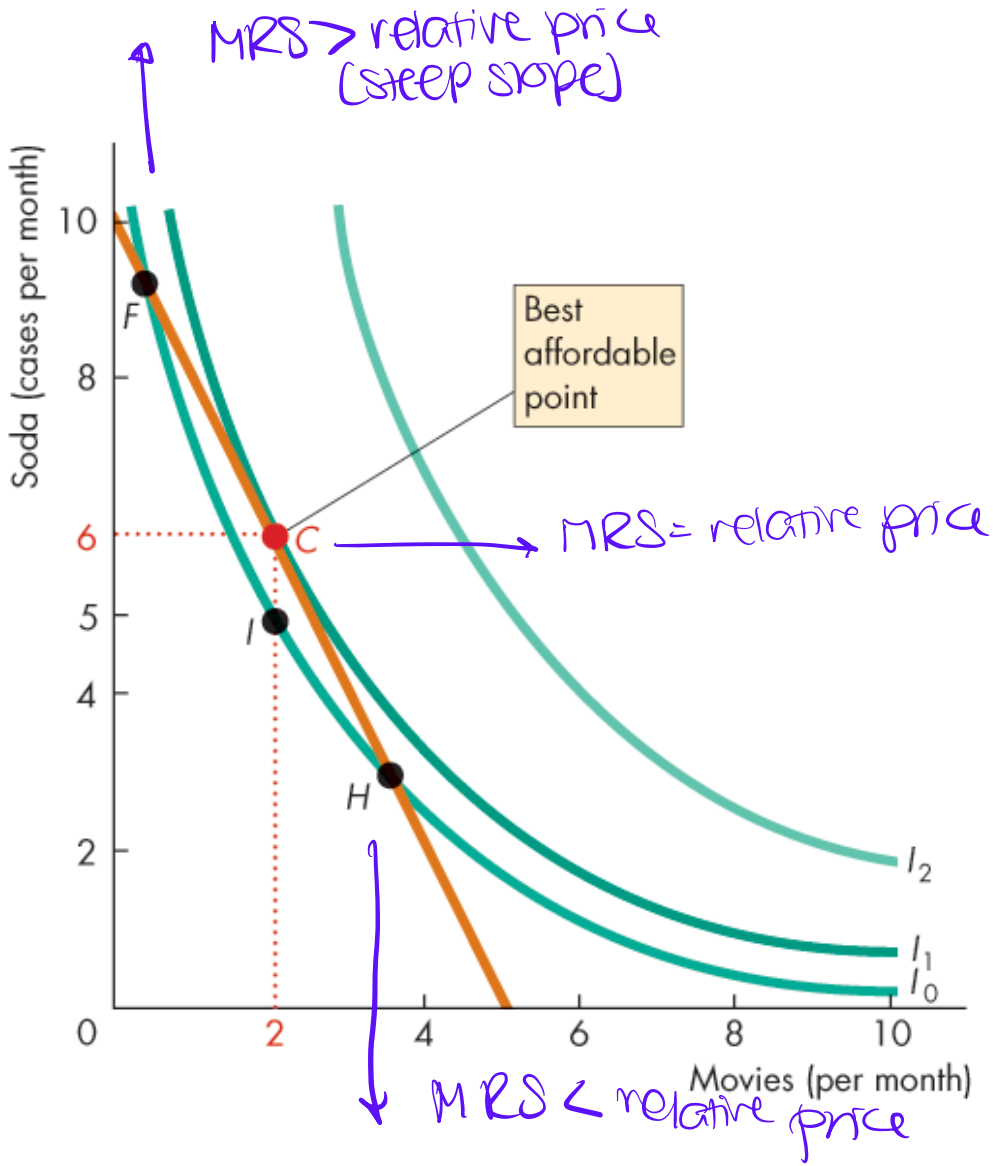
Copyright © 2023 Pearson Education, Ltd.

Predicting Consumer Choices

At point *F*, Lisa's *MRS* is *greater* than the relative price.

At point *H*, Lisa's *MRS* is *less* than the relative price.

At point *C*, Lisa's *MRS* is *equal to* the relative price P_M/P_S.



Copyright © 2023 Pearson Education, Ltd.

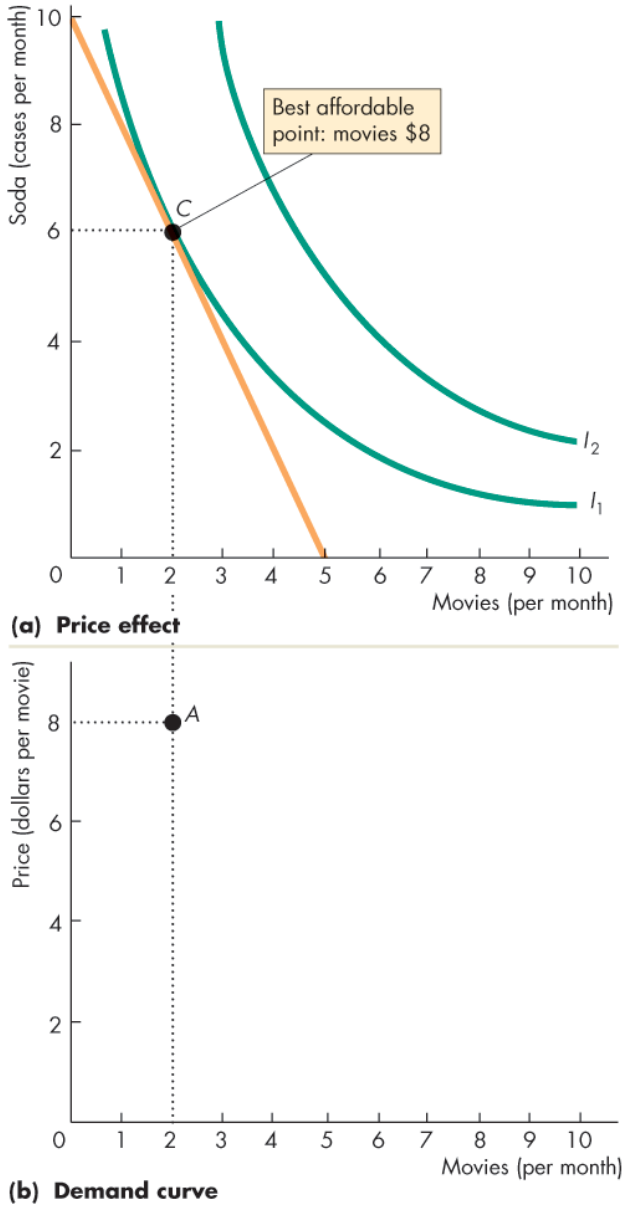
Predicting …

A Change in Price

The effect of a change in the price of a good on the quantity of the good consumed is called the **price effect**.

Figure 9.7 illustrates the price effect and shows how the consumer's demand curve is generated.

Initially, the price of a movie is $8 and Lisa consumes at point *C* in part (a) and at point *A* in part (b).



Copyright © 2023 Pearson Education, Ltd.

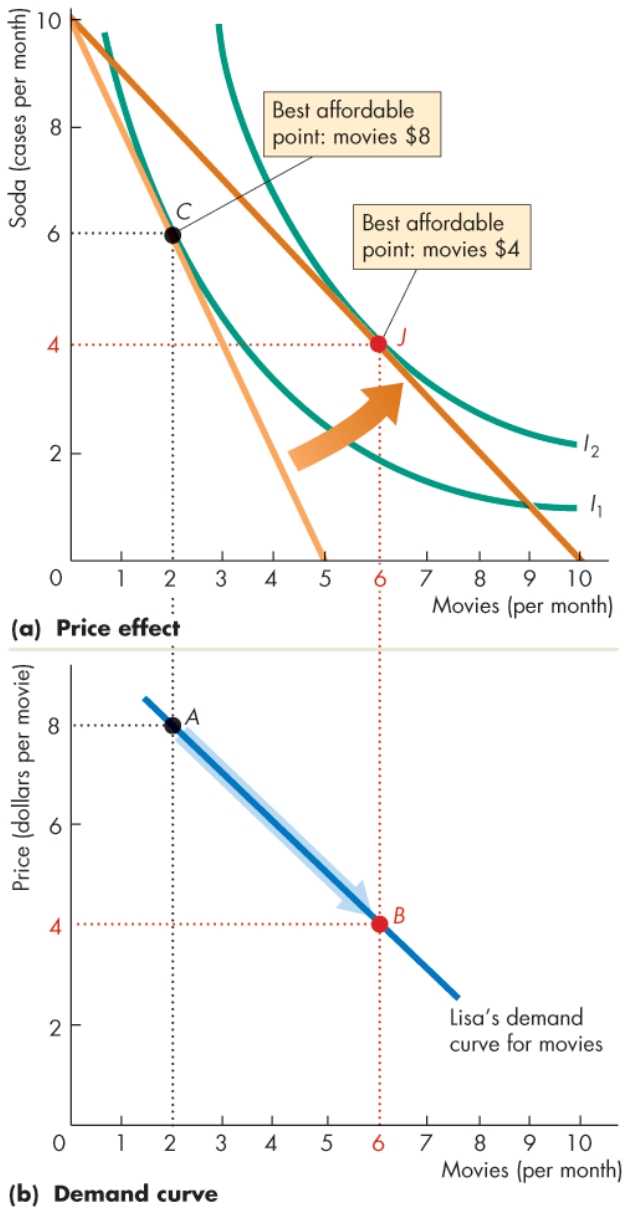
Predicting …

Now the price of a movie falls to $4.

The budget line rotates outward.

Lisa's best affordable point is now *J* in part (a).

In part (b), Lisa moves to point *B*, which is a movement along her demand curve for movies.



Copyright © 2023 Pearson Education, Ltd.

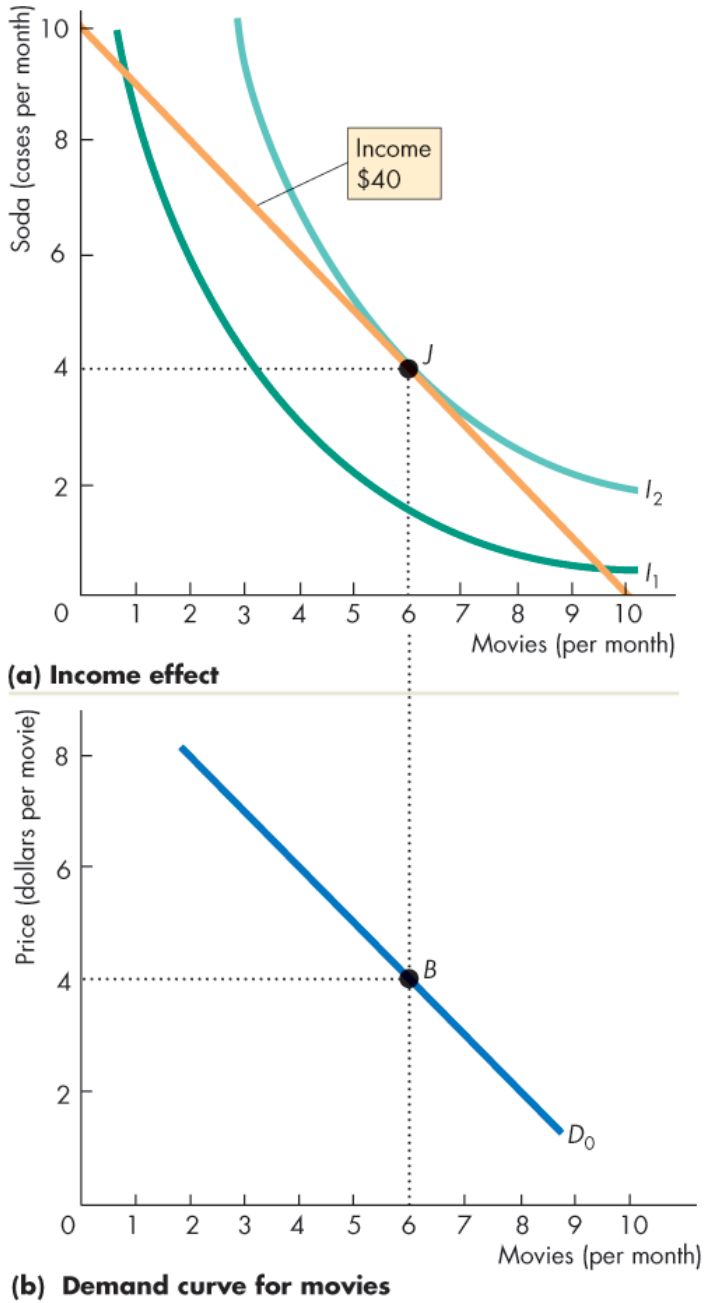
Predicting …

A Change in Income

The effect of a change in income on buying plans is called the **income effect**.

Figure 9.8 illustrates the effect of a decrease in Lisa's income with no change in the prices.

Initially, Lisa consumes at point *J* in part (a) and at point *B* on demand curve D_0 in part (b).



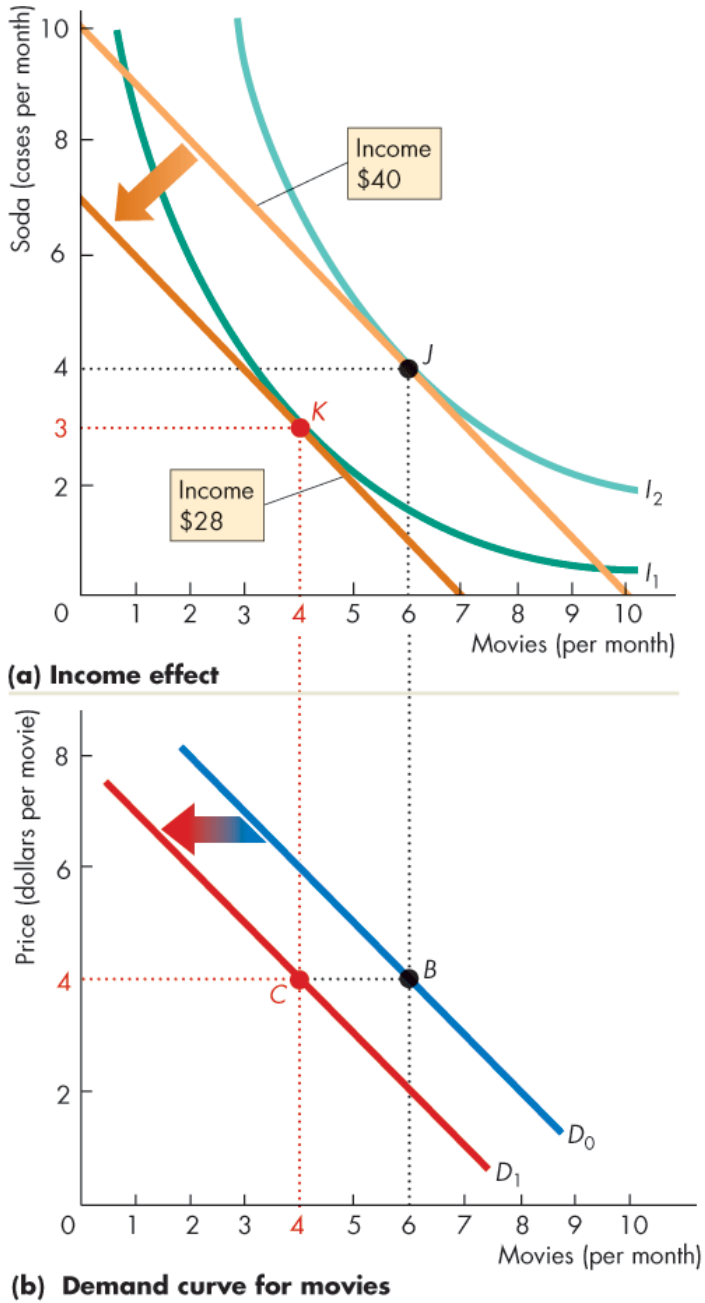
Copyright © 2023 Pearson Education, Ltd.

Predicting …

When Lisa's income decreases, her budget line shifts leftward in part (a).

Her new best affordable point is *K* in part (a).

Her demand for movies decreases, shown by a leftward shift of her demand curve for movies in part (b).



Copyright © 2023 Pearson Education, Ltd.

Predicting Consumer Choices

Substitution Effect and Income Effect

For a normal good, a fall in price *always* increases the quantity consumed.

We can prove this assertion by dividing the price effect in two parts:

- Substitution effect
- Income effect

Copyright © 2023 Pearson Education, Ltd.

Predicting Consumer Choices

Initially, Lisa has an income of $40, the price of a movie is $8, and she consumes at point *C*.

The price of a movie falls from $8 to $4 and her budget line rotates outward.

Lisa's best affordable point is then *J*.

The move from point *C* to point *J* is the **price effect**.



(a) Price effect

Copyright © 2023 Pearson Education, Ltd.

Predicting Consumer Choices

We're going to break the move from point *C* to point *J* into two parts:

price effect

1. The substitution effect
2. The income effect



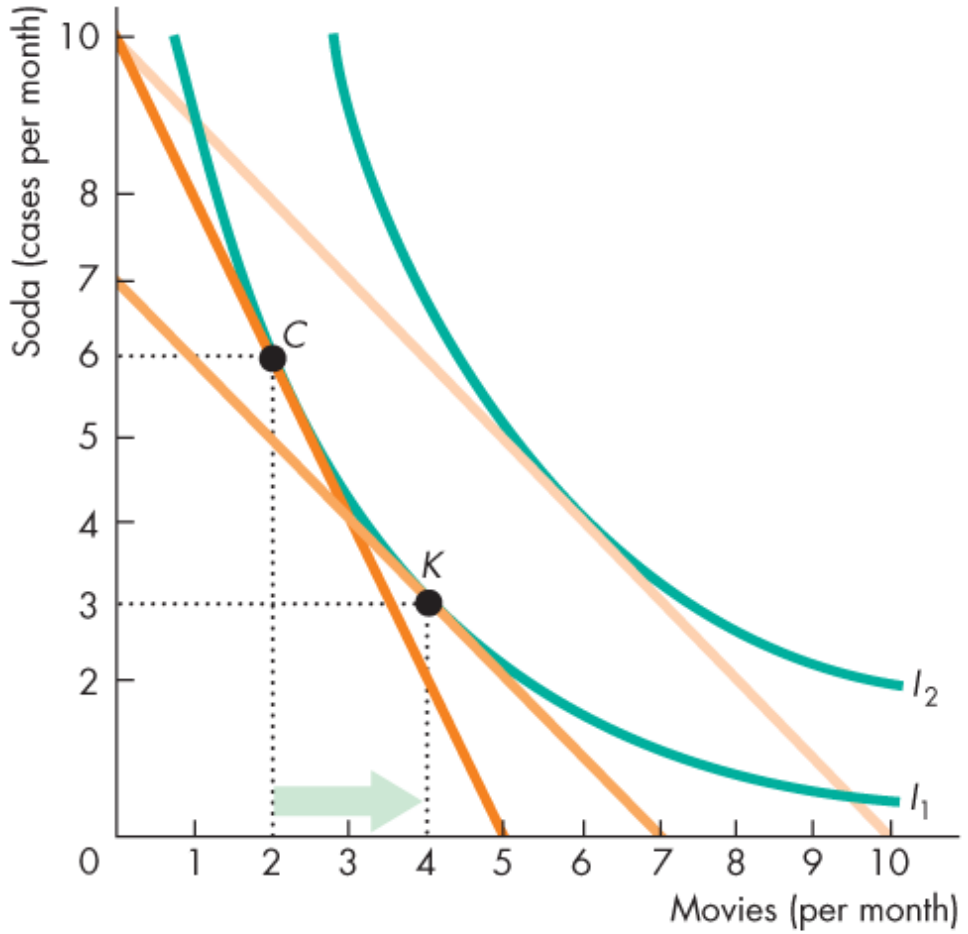
(a) Price effect

Copyright © 2023 Pearson Education, Ltd.

Predicting Consumer Choices

Substitution Effect

The **substitution effect** is the effect of the change in price on the quantity bought when the consumer *remains on the same indifference curve*.



(b) Substitution effect

Copyright © 2023 Pearson Education, Ltd.

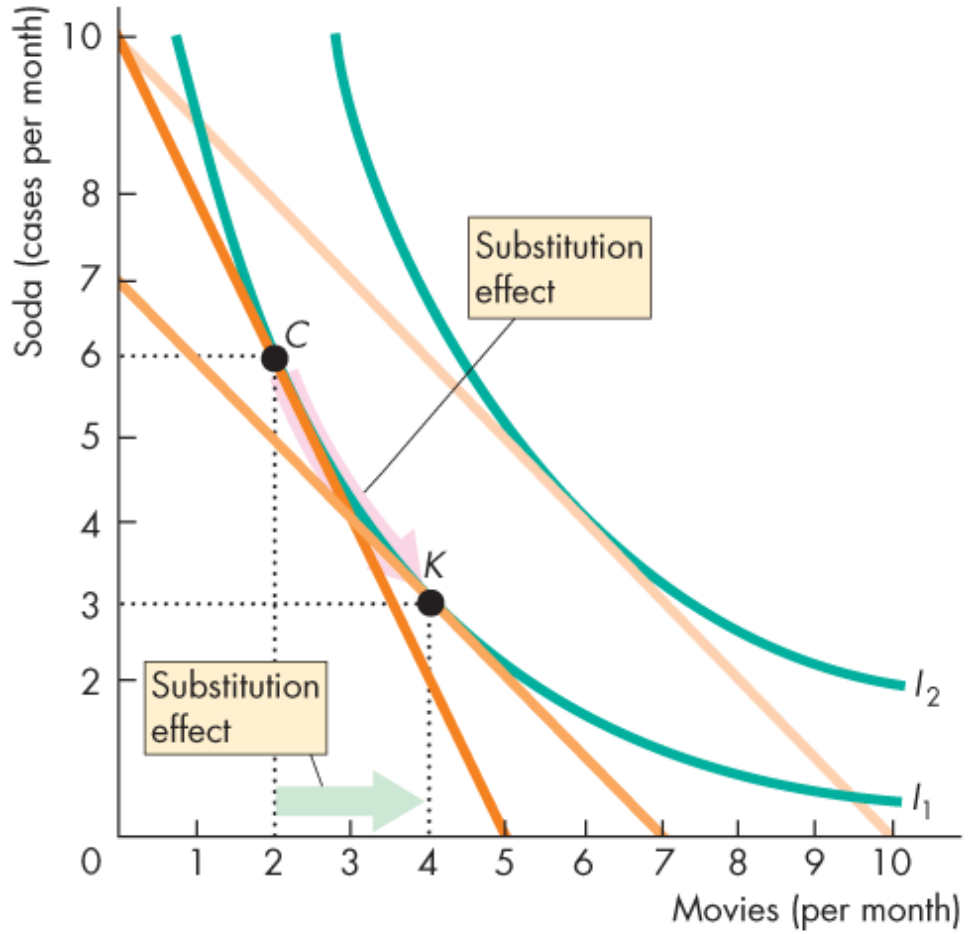
Predicting Consumer Choices

To isolate the substitution effect, we give Lisa a hypothetical pay cut.

Lisa is now back on her original indifference curve but with the price of a movie at $4.

Her best affordable point is *K*.

The move from *C* to *K* is the **substitution effect**.



(b) Substitution effect

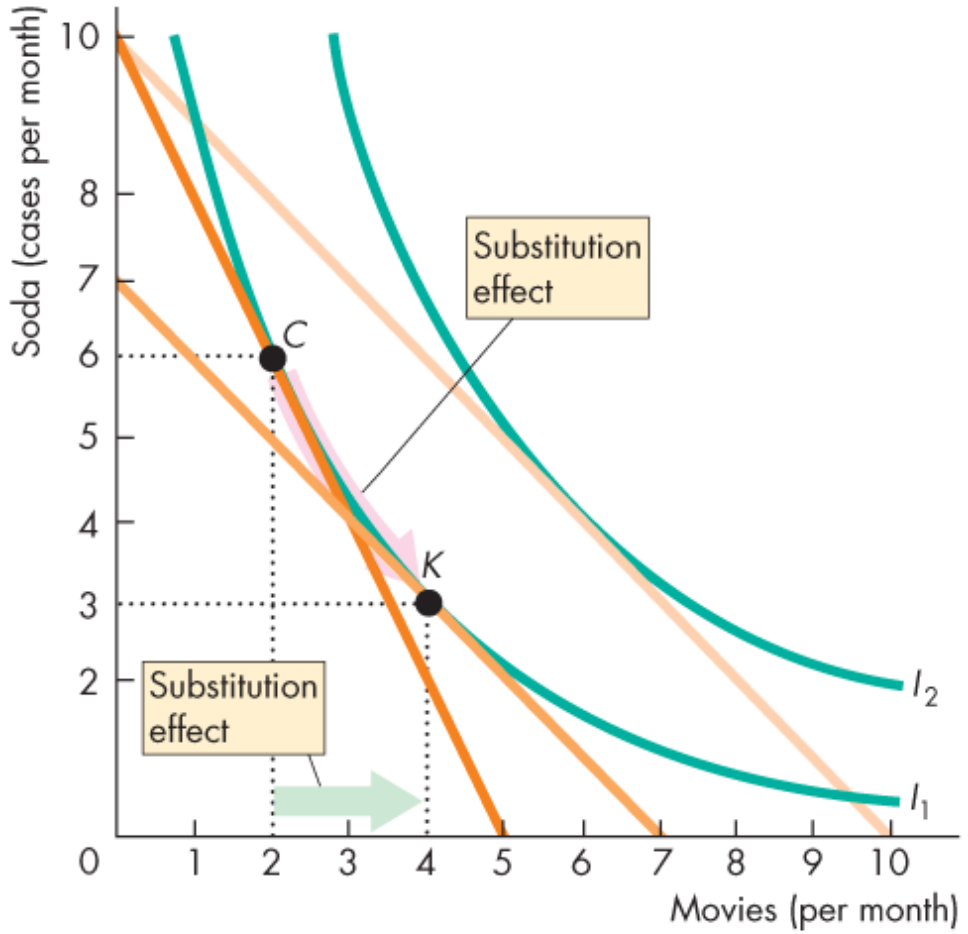
Copyright © 2023 Pearson Education, Ltd.

Predicting Consumer Choices

The direction of the substitution effect never varies:

When the relative price falls, the consumer always substitutes more of that good for other goods.

The substitution effect is the first reason why the demand curve slopes downward.



(b) Substitution effect

Copyright © 2023 Pearson Education, Ltd.

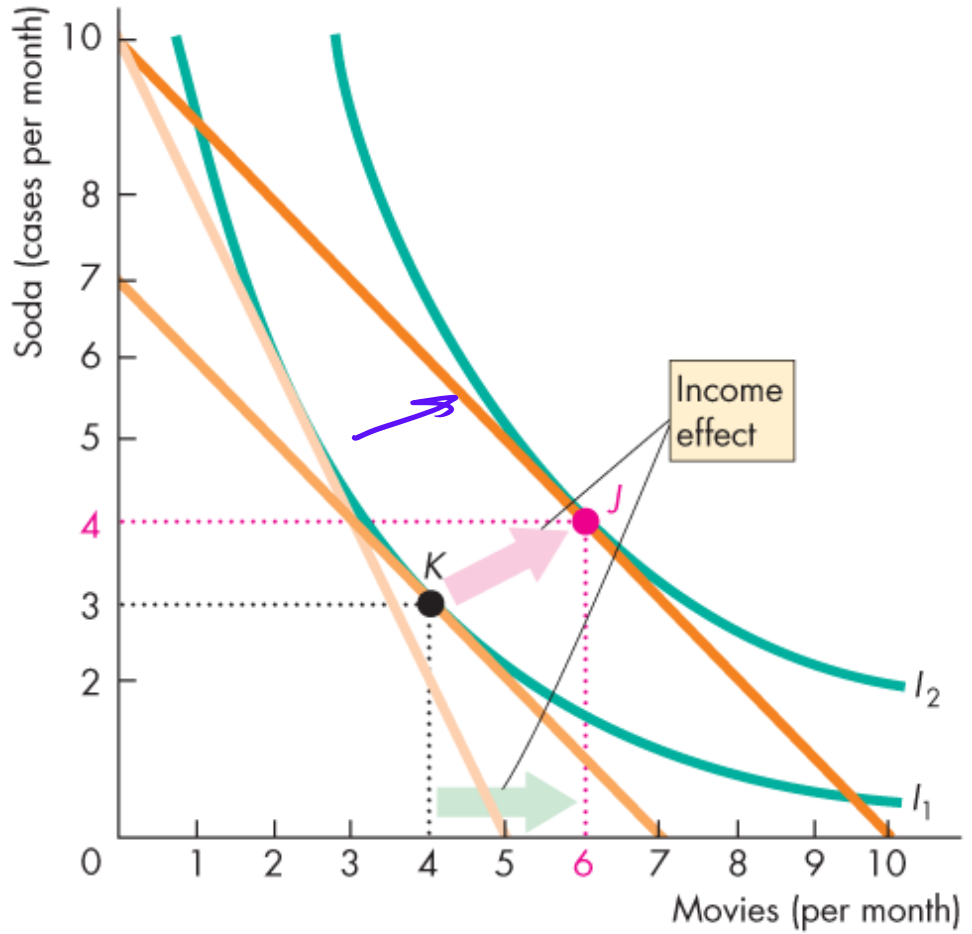
Predicting Consumer Choices

Income Effect

To isolate the income effect, we reverse the hypothetical pay cut and restore Lisa's income to its original level (its actual level).

Lisa is now back on indifference curve I_2 and her best affordable point is J.

The move from K to J is the income effect.



(c) Income effect

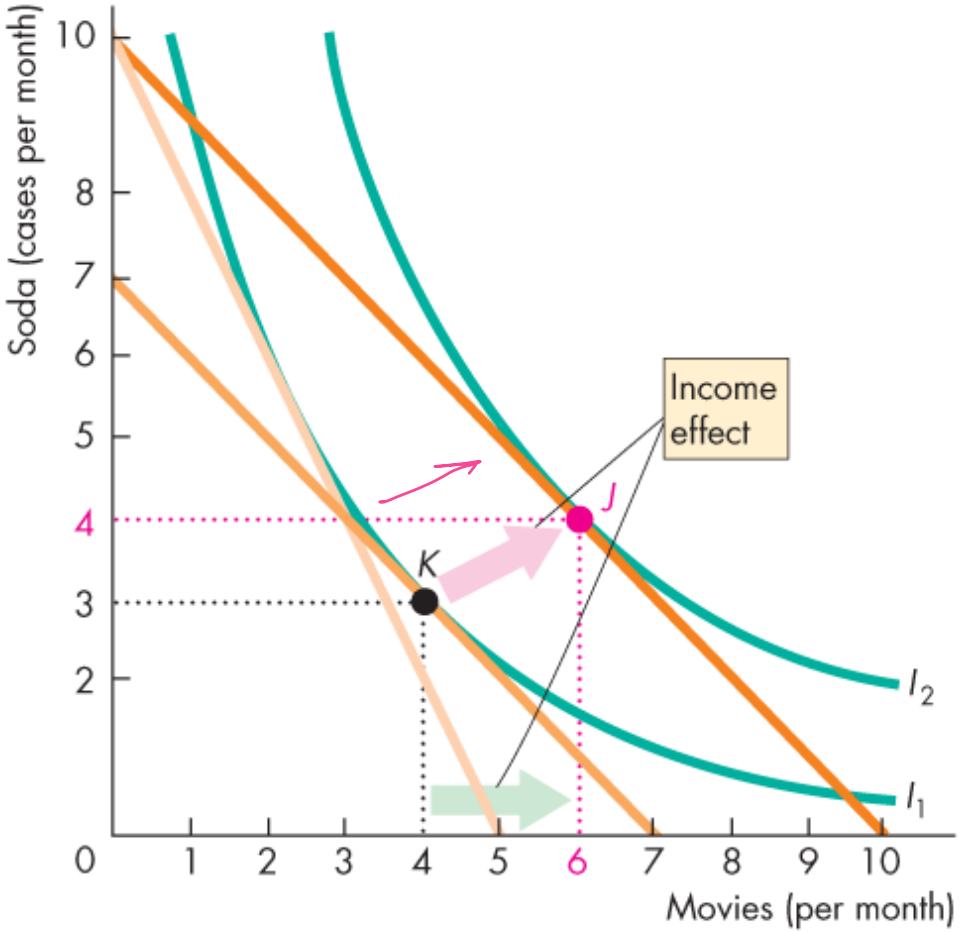
Copyright © 2023 Pearson Education, Ltd.

Predicting Consumer Choices

For Lisa, movies are a normal good.

With more income to spend, she sees more movies—the income effect is positive.

For a normal good, the income effect *reinforces* the substitution effect and is the second reason why the demand curve slopes downward.



(c) Income effect

when price falls, more quantities are bought increasing purchasing power

Copyright © 2023 Pearson Education, Ltd.

and vice versa

Predicting Consumer Choices

Inferior Goods

For an inferior good, when income increases, the quantity bought decreases.

The income effect is negative and works against the substitution effect.

As long as the substitution effect dominates, the demand curve still slopes downward.

normal goods → positive income effect → reinforces substitution effect
inferior goods → negative income effect → works against substitution

Copyright © 2023 Pearson Education, Ltd.

Predicting Consumer Choices

If the negative income effect is stronger than the substitution effect, a lower price for inferior goods brings a *decrease* in the quantity demanded—the demand curve slopes upward!

This case does not appear to occur in the real world.

Back to the Facts

The best affordable choices determine spending patterns.

Changes in prices and incomes change the best affordable point and change consumption patterns.

Copyright © 2023 Pearson Education, Ltd.